

RADAR DEMO IN C

■ MANUJ ARORA

Radar (short for 'Radio Detection And Ranging') is used to detect and track distant objects (targets). It can identify the range, altitude, direction or speed of both moving and fixed objects such as aircrafts, ships, spacecrafts, mountain ranges, radio and TV towers, and guided missiles.

Basic principle

Radars make use of the echo phenomenon. The system utilises the fact that electromagnetic waves reflect off objects. The radar dish or antenna transmits pulses of radio waves or microwaves.

A typical radar antenna is shown in Fig. 1. If a radio wave encounters sudden change in conductivity, permittivity or permeability in the medium, a part of the electromagnetic energy gets absorbed by the object and is reradiated. This sudden change in the electrical properties of the medium then indicates the target. This reradiated energy, on being received back at the radar station, gives information about the location of the target.

As an example, if a ship's whistle is sounded in the middle of the ocean, the sound waves will dissipate their energy as they travel outward and at some point disappear entirely. However, if the whistle is sounded near an object such as a cliff, some of the radiated sound waves will reflect back to the ship as an echo.

Distance determination. The distance of the target from the radar is determined from the running time of the high-frequency transmitted signal and the propagation velocity (C). The actual range of a target from the radar is known as slant range (R). Since the waves travel to a target and back, the round-trip time (T) is divided by '2' in



Fig. 1: Typical radar antenna

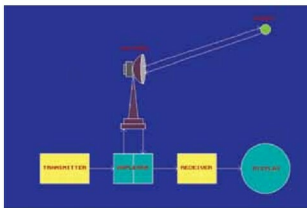


Fig. 2: Block diagram of radar communication system



Fig. 3: When the antenna starts radiating the RF waves towards the target, the timer turns on

order to obtain the time that the wave took to reach the target. Thus the slant range can be calculated as:

$$R = \frac{C \times T}{2}$$

Direction determination. The angular direction of the target is determined by the directivity of the antenna. Directivity is the ability of the antenna to concentrate the transmitted energy in a particular direction. By measuring the direction in which the antenna is pointing when the echo is received, both the azimuth and elevation angles from the radar to the object can be determined. The accuracy of angular measurement is determined by the directivity, which is a function of the size of the antenna.

Software program

The above concept has been taken into consideration while developing the program. The program is written in 'C' language and compiled using Turbo C IDE. Turbo C has a good collection of graphics libraries.

This program uses various C graphics files such as `egavga`, `bgi`, `scri.chr`, `trip.chr` and `tscr.chr`, and can be divided into three parts.

First part. This part of the program demonstrates the basic echo principle of the radar that involves reflection of waves from the target back to the radar set that is used to locate the

